

A small supervision budget barely improves weak-to-strong generalization

On GPT-2 / BoolQ, mixing in a ground-truth budget helps only modestly — and *where* and *how* it's spent show little effect.

A consistent picture emerges from the data: the student largely imitates the teacher's errors, and ground truth appears to repair mainly what it directly supervises — pointing to *volume* as the main lever.

GPT-2 family · BoolQ + SciQ · 3 seeds · paired per-(pair,seed) contrasts · pre-registered

Scope & setup

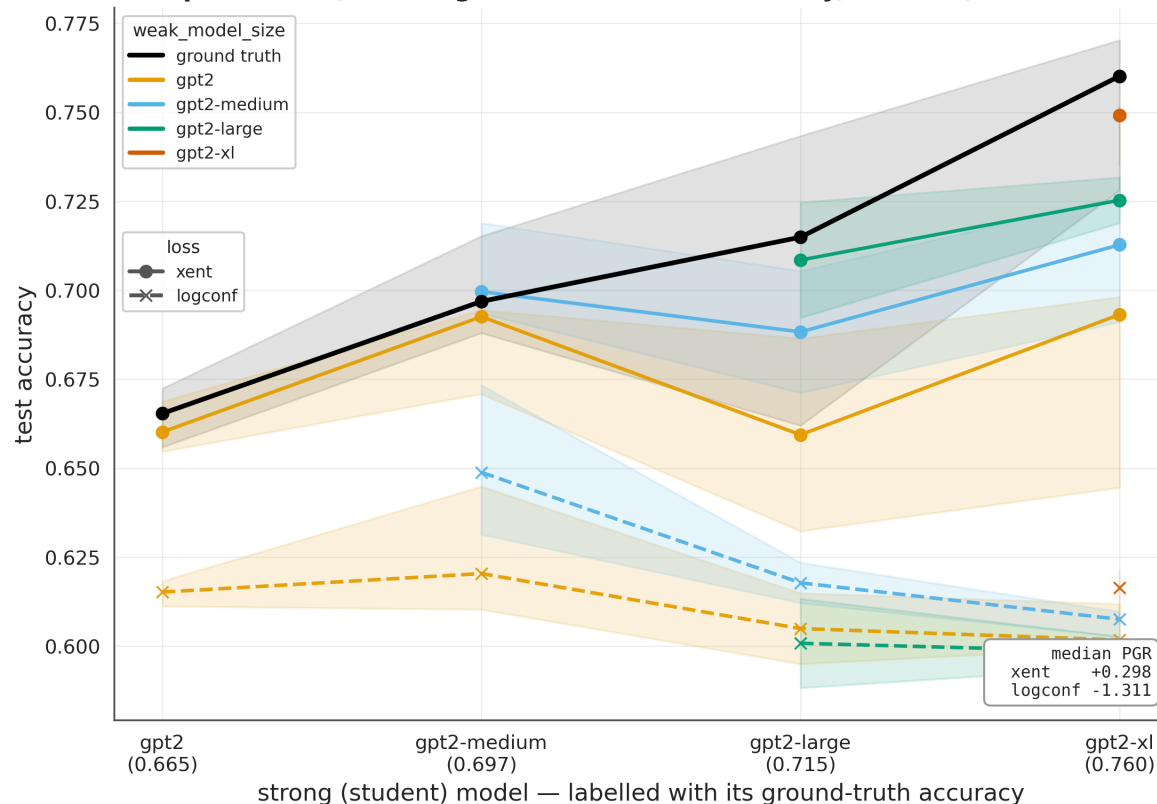
- **Setting:** weak-to-strong generalization — a strong *student* is trained on a weak *teacher's* labels.
- **Extension:** a *supervision budget* — mix a fraction of ground-truth ("strong") labels into the weak labels, and sweep that fraction.
- **Models: GPT-2 family only** — gpt2 / medium / large / xl, with within-family student-teacher pairs.
- **Tasks: BoolQ** (required) + **SciQ** (cross-task check).
- **Readout:** median **PGR** across the model sweep (raw accuracy primary; PGR secondary).

3 seeds · paired per-(pair,seed) contrasts · pre-registered · gpt2-large (seed 1) excluded by rule

GPT-2 sweep at 25% ground truth

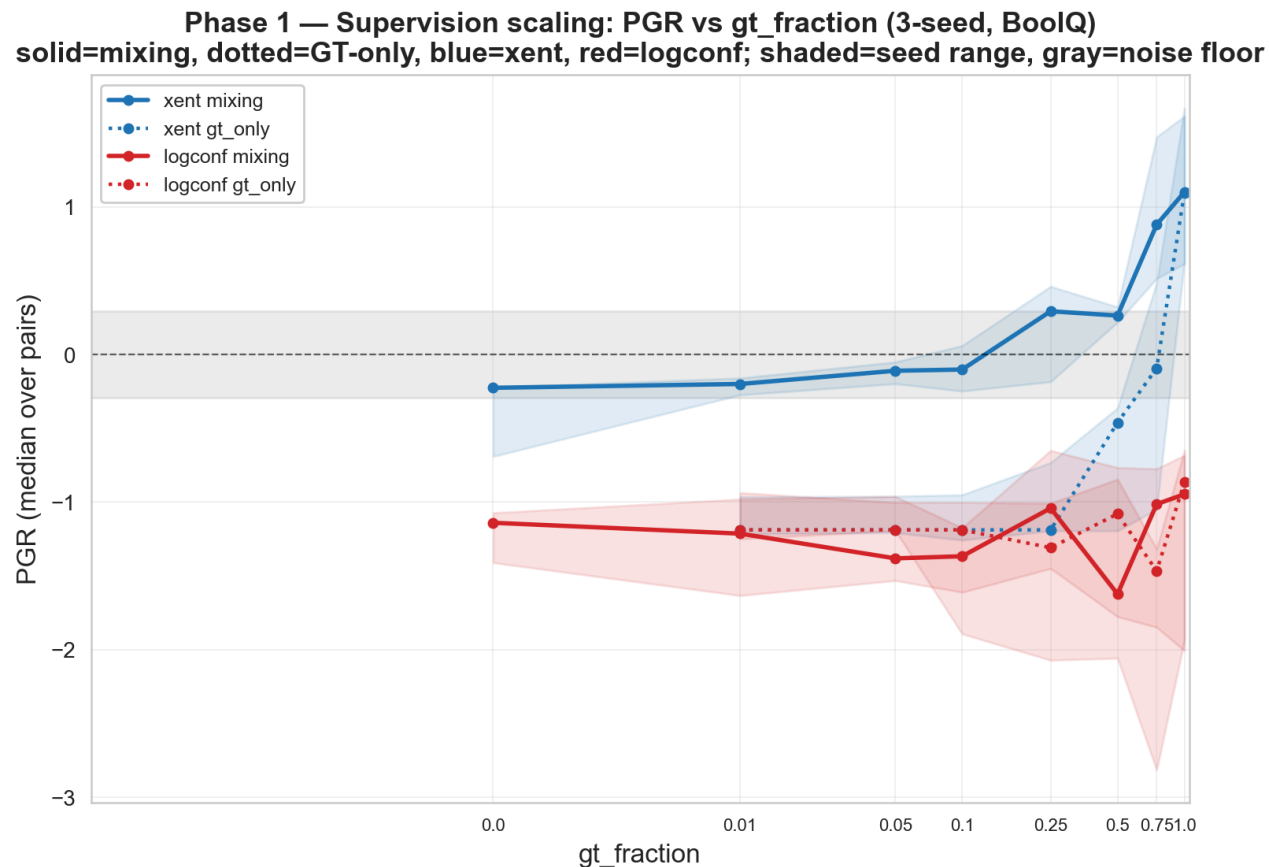
- GPT-2 family, BoolQ, **25% ground truth mixed into the weak labels** — paper-format sweep.
- **Median PGR (xent) = +0.30** across the sweep; logconf deeply negative (-1.3), so it's dropped.
- 25% GT moves the median from **-0.27 (0% GT) → +0.30**: real, but modest.
- Rest of the talk: *how* modest, where/how it does or doesn't help, and why.

W2SG sweep — BOOLQ + 25% ground truth (GPT-2 family, 3 seeds, band = seed range)



How *much*? — no cheap knee; saturates by ~75%

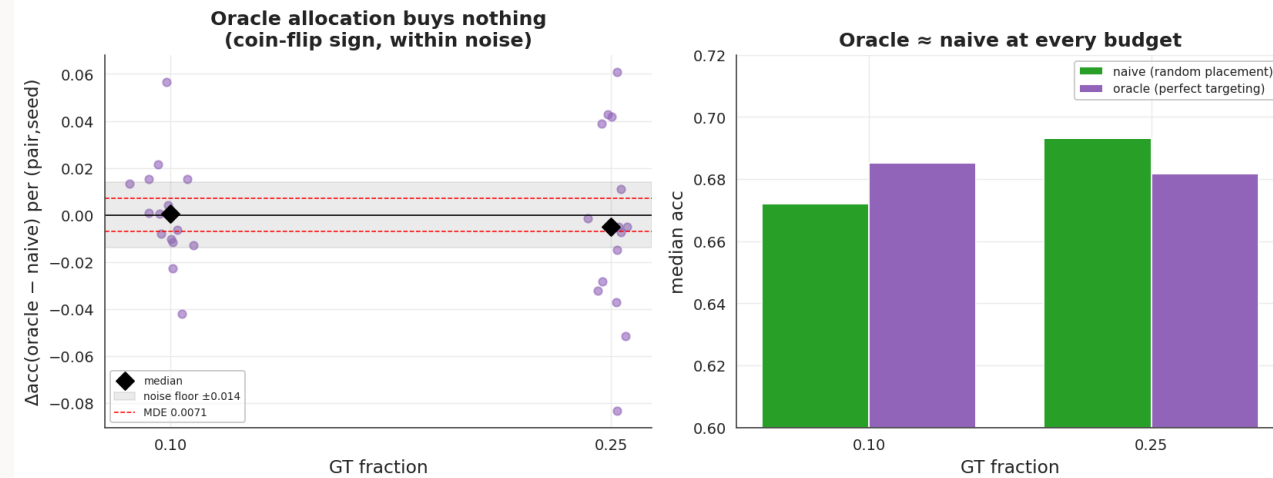
- $\leq 10\%$ GT sits within the noise floor (0.014) — a small budget adds little above the pure-weak baseline.
- The response is **back-loaded**. Median xent PGR:
-0.22 $\rightarrow$ +0.30 (0.25) $\rightarrow$ +0.28 (0.50) $\rightarrow$ +0.90 (0.75) $\rightarrow$ +1.04 (1.0).
- The 0.75 point shows it **saturates by ~75%** — the 0.75 $\rightarrow$ 1.0 step is within noise.
- A pre-registered "concave knee at 25%" prediction was **refuted** (retracted after the multi-seed view).



Where it's spent — little effect (allocation null)

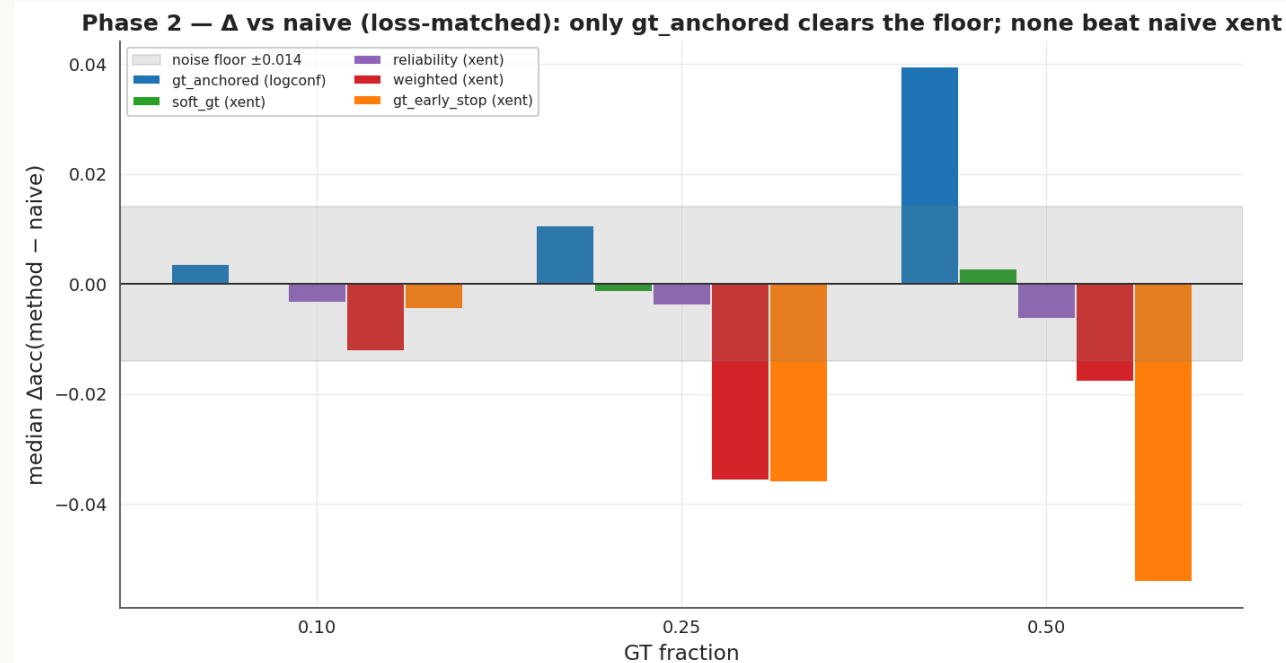
- A **perfect error-targeting oracle** places GT exactly on the teacher-wrong rows — an upper bound on any allocation heuristic.
- oracle - naive (random placement) = **+0.0006 / -0.0049** at 0.10 / 0.25 — inside the MDE (0.0071).
- Since even this oracle ties random placement, allocation heuristics have **little room to help here**.

Phase 1b · B — WHERE you spend GT does not matter (allocation NULL): a perfect error-targeting oracle ties random



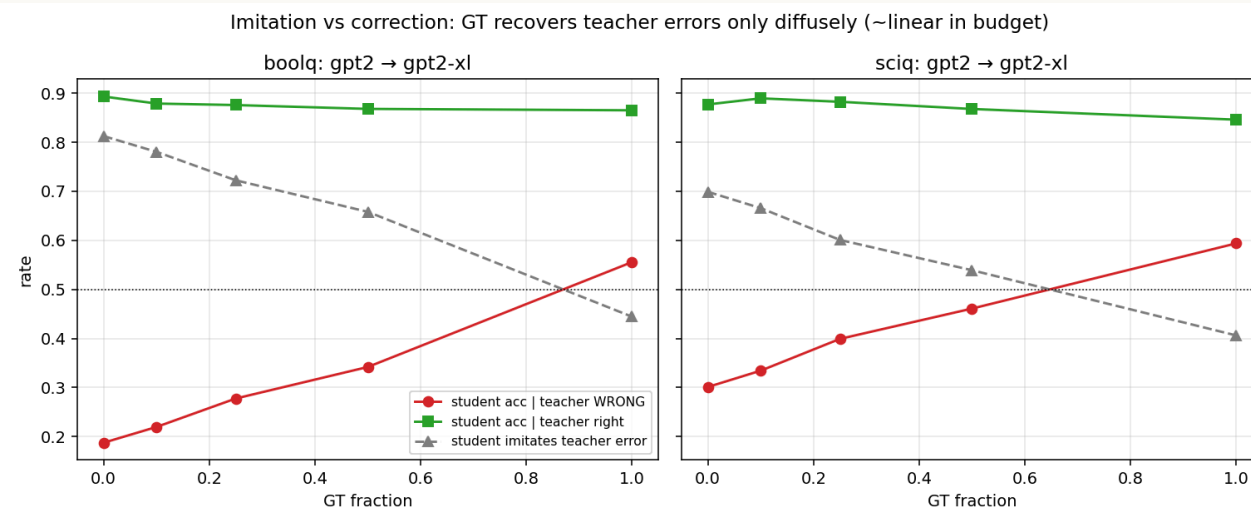
How it's combined – little effect (combination null)

- A pre-registered portfolio of **5 methods**: GT up-weighting, soft-GT, GT-anchored logconf, reliability-weighting, GT-early-stop.
- **None** left-shifts the curve or raises the ceiling – median Δ vs naive mixing ≈ 0 .
- Only **gt-anchored** clears the floor (+0.040 @0.50), and it only *rescues* logconf – still below plain xent (0.642 vs 0.697).



Why: the student imitates the teacher's **errors**

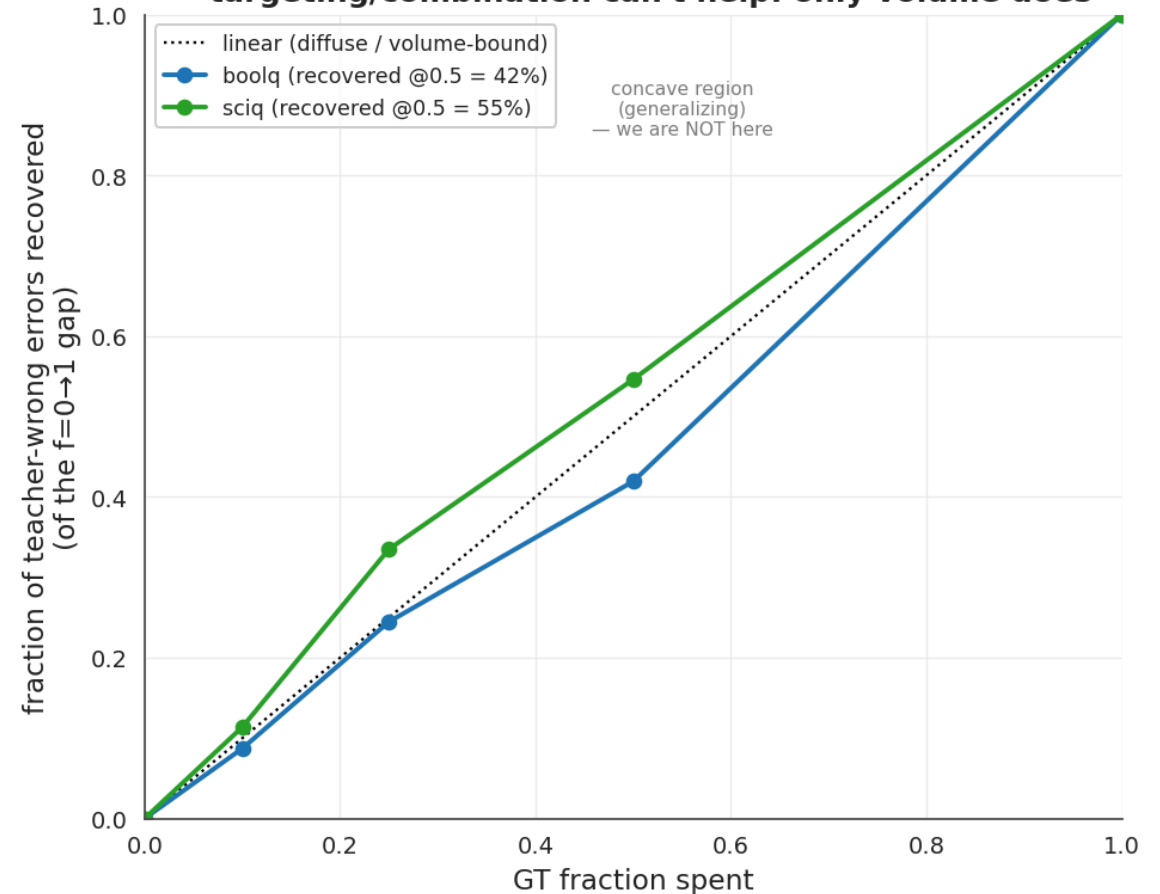
- Mechanism probe (gpt2 → gpt2-xl), on per-example predictions.
- At 0% GT, the student copies the teacher's **wrong** answer **81% (BoolQ) / 70% (SciQ)** of the time.
- The failures look **largely inherited rather than independent** — the rows needing correction are mostly the teacher-wrong ones.



Why (hypothesis): recovery looks **diffuse** → volume may be the main lever

- GT repairs teacher-wrong rows **~linearly in budget** (42% / 55% of the gap at 50%) — rather than concave/targeted.
- A GT label's value appears **roughly independent of which row it lands on**.
- This is **consistent with all three results**: if recovery is volume-bound, *where* (allocation) and *how* (combination) have little leverage — leaving *how much* as the main lever.

Mechanism — error recovery is ~LINEAR in budget (diffuse), not concave → targeting/combination can't help: only volume does



Are the nulls real? — power and controls

- **Power:** paired per-(pair,seed) contrasts give **MDE = 0.0071**, below the 0.02 effect of interest — a null here is informative.
- **Pre-registration:** git-anchored predictions, hits *and* misses reported; the "knee" prediction was retracted.
- **Exclusion:** an 8-seed study found gpt2-large GT collapses **27%** of the time → excluded by rule, not by outcome.
- **De-confounding:** a random-label control shows weak labels are genuinely informative (mixing > GT-only isn't just row count).
- **Cross-task:** the findings replicate on **SciQ**.

Experiments across the three axes

Axis	Tried	Outcome
How much	8-point fraction sweep, $0 \rightarrow 1$	back-loaded, saturates ~ 0.75
Where	error-targeting oracle + random control	null — no signal on allocation
How	5 combination / loss methods	null — 1 floor-clearer, still $< \text{xent}$
Loss	xent vs logconf	logconf inert $\rightarrow$ dropped
Tasks	BoolQ + SciQ	replicates on both
Mechanism	imitation-vs-correction probe	consistent with the nulls
Robustness	8-seed variance, seeds 3–4 reserve	exclusion principled

Most results are negative; together they point toward a common mechanism rather than leaving loose ends.

Takeaways & the experiment that would change this

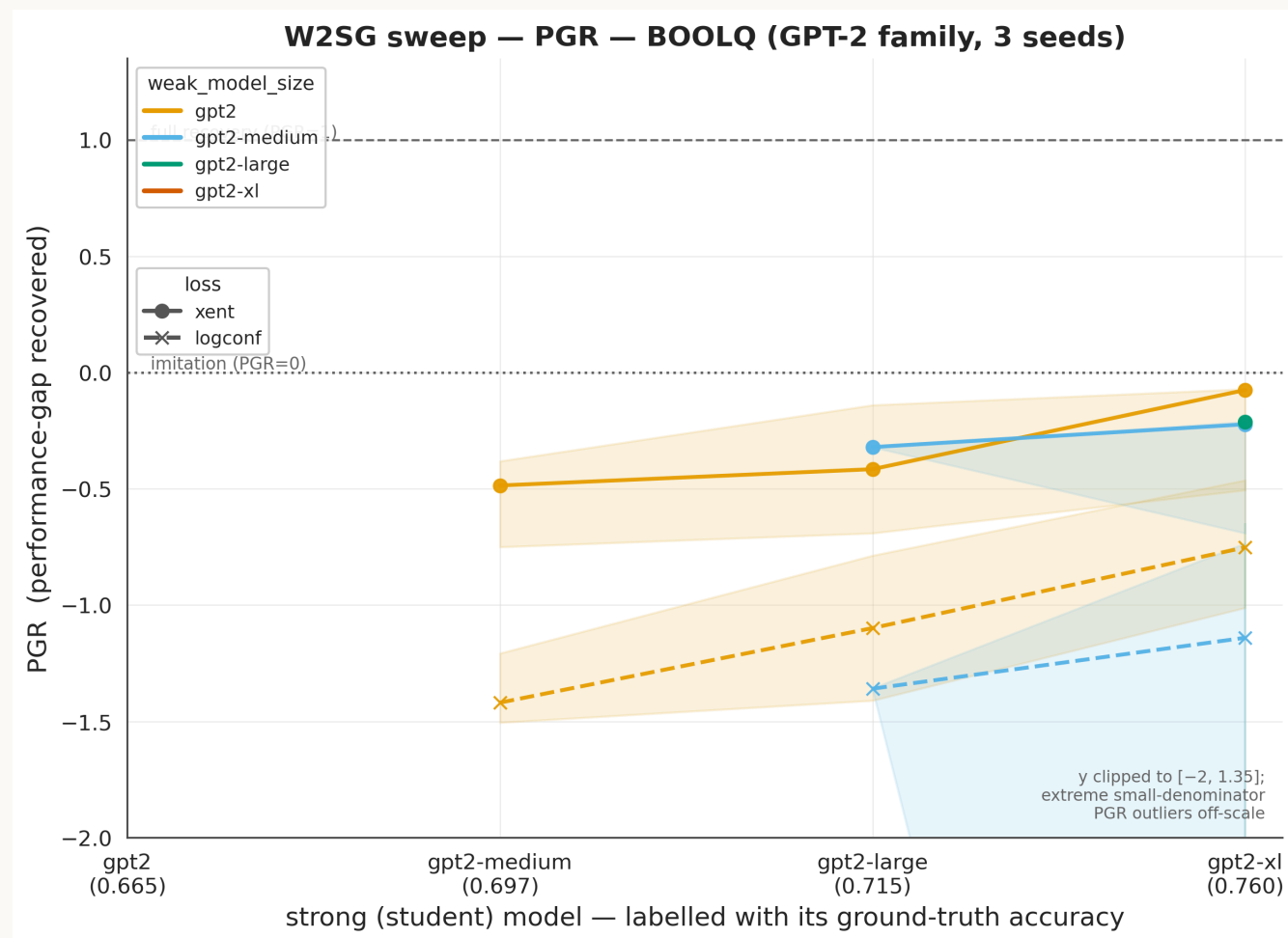
- Reproduced W2SG on the GPT-2 family and extended it to a supervision-budget setting.
- A **three-question decomposition** — *how much* / *where* / *how* — with a powered null on each.
- A single **hypothesis** (imitation + volume-bound recovery) is consistent with all three.
- **The data point to scale — rather than allocation or combination — as the main constraint here.**
 - The test that could change this: **larger student-teacher capacity gaps** (out of scope here — GPT-2 only).
 - Prediction: if recovery becomes **concave** at scale, *where* and *how* begin to matter.

Appendix

frac=0 reproduction (acc + PGR) · transfer heatmap · per-method × pair detail · de-confound · cross-task

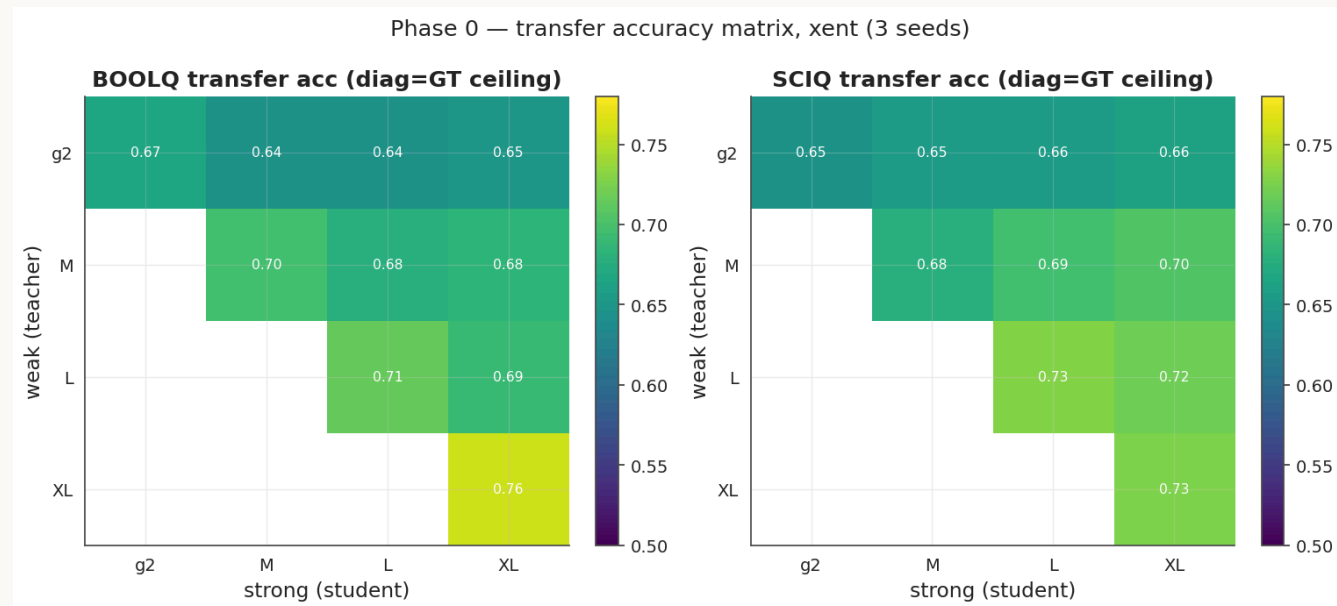
Appendix — frac=0 reproduction (the testbed)

- Canonical W2SG sweep, GPT-2 family, BoolQ, **0% GT** — PGR axis.
- Median xent PGR **-0.27**: BoolQ is low-signal at zero strong supervision (several pairs sub-imitation).
- The baseline the 25%-GT plot is measured against.



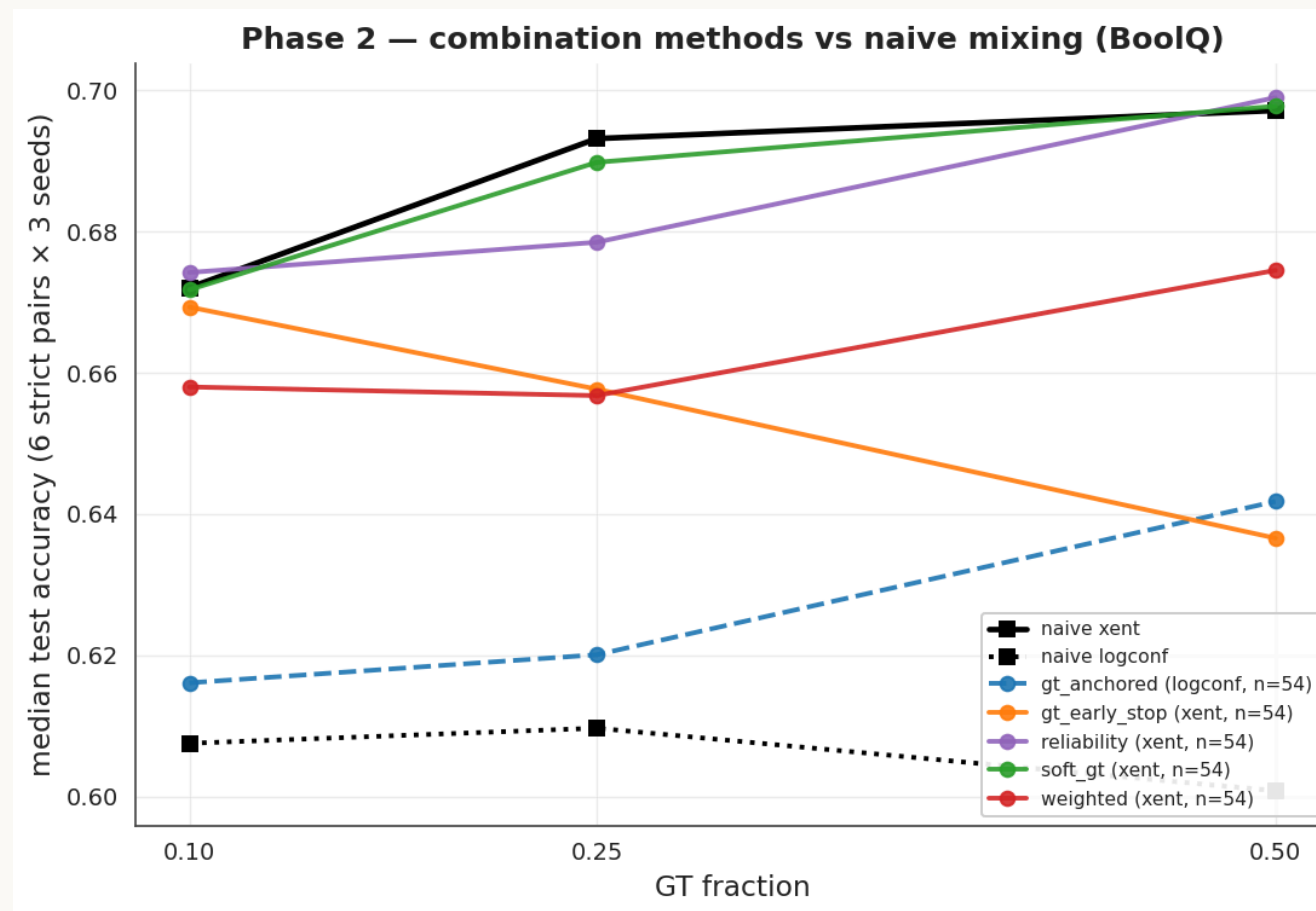
Appendix — transfer tracks the teacher more than student capacity

- Holding the teacher fixed, scaling the student up adds $\sim +0.003$; improving the teacher adds $\sim +0.04$ ($\sim 10\times$ here).
- Suggests the student's extra capacity is largely spent on imitation — the static counterpart of the mechanism.



Appendix – combination portfolio, per-method detail

- All five methods track naive mixing across $\{0.10, 0.25, 0.50\}$.
- gt-anchored (logconf) is the only floor-clearer; it rescues logconf but stays under xent.



Appendix — weak labels are informative (de-confound)

- Ordering: **random** < **gt_only** < **naive mixing** (15/15 BoolQ, 18/18 SciQ pairs).
- Replacing weak labels with noise *hurts* → mixing's gain reflects weak-label information, not just training-set size.

